Graphic Era deemed to be University DEHRADUN

PROJECT AND TEAM INFORMATION

Project Title

EduBridge- A Role-Based Academic Collaboration and Management Platform

Student/Team Information

Team Name: Team ID- FS -T082	TriConnect
Team member 1 (Team Lead)	Chaturvedi, Shraddha – 23022210 chaturvedishraddha73@gmail.com

Team member 2	Agarwal, Gauri –23022497 gauriagarwal0805@gmail.com
Team member 3	Guliyani, Gauri- 230221455 guliyanigauri@gmail.com

PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract

EduBridge is a web-based platform used to manage academic activities and improve communication between students, faculty, and management in universities and colleges. It helps with many important academic functions together in one place, so users don't have to rely on multiple platforms. This helps in making the system more organized and efficient.

The platform is built using a three-tier architecture. Frontend is built using React.js, Backend using Node.js and Express.js, and database using PostgreSQL. There are three main types of users in the system- students, faculty, and management, and each of them gets separate dashboards with features based on their role.
Some of the main features of EduBridge are:

- A faculty recommendation system that helps students find teachers based on their expertise and research areas.
- There is also a real-time messaging system that allows direct communication between students and faculty.
- The platform has a feedback system where students can rate teachers and write comments.
- There is a notice board for sharing announcements with specific groups, timetable management with file upload options, and course material sharing.

The system uses JWT-based authentication to control access. There is also Role-based password authentication. The backend includes 50+ RESTful API endpoints that manage different functionalities like user messaging, feedback, recommendations, notices, courses, and timetables.

The system uses 13 database tables which are structured with normalization, foreign key relationships to ensure data integrity. There is also a sentimental analysis, which will use Python and VADER to analyze student feedback, but we are still working on this part.

Updated Project Approach and Architecture

The project is build using a simple three-layer structure that includes frontend, backend, and database.

First, the frontend is developed using React, where users can interact with the system through different pages like login, dashboard, messaging, and notice board. It provides a clean and easy interface for students, faculty, and admin.

Second, the backend is built using Node.js and Express. It handles all the main logic such as user authentication, sending messages, posting notices, managing feedback, and processing requests from the frontend.

The frontend and backend communicate through REST APIs, where data is sent and received in JSON format. JWT is used for secure login and role-based access so that each user can only access their own features.

Finally, PostgreSQL is used as the database to store all the information such as users, messages, feedback, and timetables.

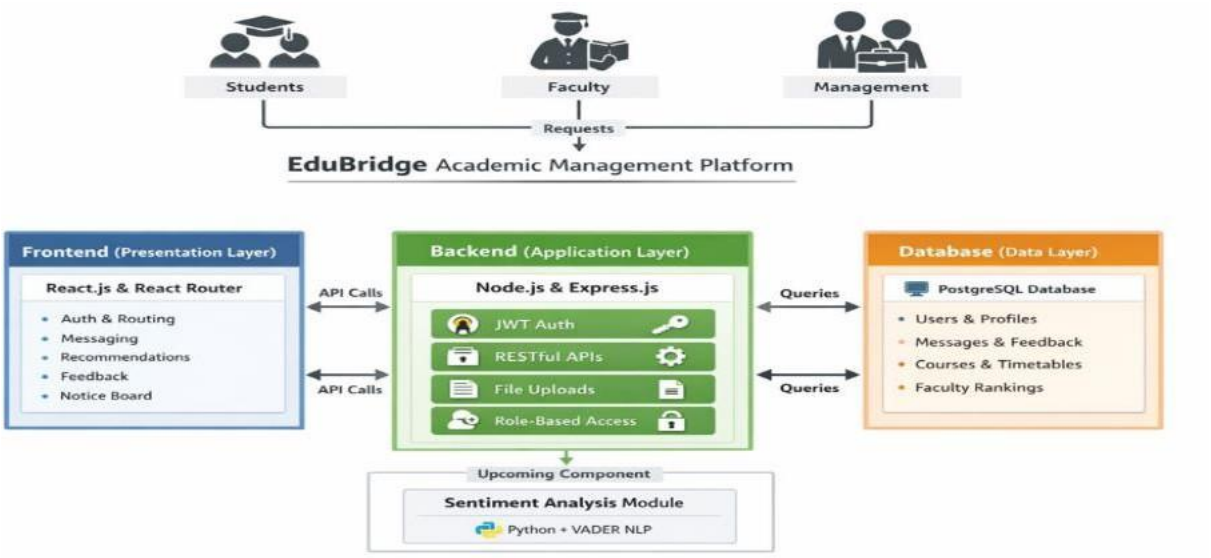

FIGURE 1: Updated System Architecture of EduBridge

Tasks Completed

Task Completed	Team Member
Database & Backend Development: • Designed database and created required tables • Developed backend APIs for core features	Shraddha
Frontend Development: • Built frontend pages using React • Designed responsive UI with custom CSS	All
Feature & Integration: • Created messaging interface and feedback submission forms	Gauri Agarwal+ Gauri Guilyani
• Implemented login system and authentication	Shraddha + Gauri Guliyani
• Built notice board and analytics dashboard • Performed basic testing of features	Gauri Agarwal
• Sentiment Analysis Integration • Create feedback and update Faculty Management page to display all student feedback. • Improve UI design and responsiveness • Perform full testing of all features • Fix remaining bugs and errors • Setup deployment for final project	All (Team work)

Challenges/Roadblocks

During the project, we faced some challenges.
One major problem was designing the database properly. In the beginning, some column names did not match with the backend code, which caused errors. We fixed this by correcting the schema and updating the queries.

Another challenge was managing the login session. Sometimes users were logged out suddenly. We solved this by handling errors properly and managing tokens carefully.

We also had problem in feedback section. When student gave feedback to any teacher it was not visible to admin and we have resolved it by correcting the tables in database and writing the correct backend logic.

The main issue we faced was difficulty in making all features work together smoothly.

Overall, we solved these problems by testing step by step and discussing as a team.

Future Scope

While the core project is complete and fully functional, the following enhancements can be implemented in future versions:

1. Real-Time Communication Features:

- Live notifications for announcements
- Data encryption possible to ensure data is encrypted.
- Online status indicators for faculty availability

2. Advanced Analytics and Reporting:

- Faculty performance tracking and engagement metrics
- Sentiment trend analysis over time

3. Mobile Application Development:

- Native mobile apps for iOS and Android
- Push notifications for updates
- Offline mode support

Project Outcome/Deliverables

The outcome of EduBridge project is a web-based platform that brings students, faculty, and management together on a single system for better communication and management.

The system consists of a frontend interface built using React, where users can log in and access features based on their role. Students will be able to view notices, check timetables, message faculty, and give feedback. Faculty will be able to post notices, upload timetables, and view feedback given by students. Admin will be able to manage users and monitor overall activity through a dashboard.

The frontend communicates with the backend, which is using Node.js and Express. The backend handles all the main operations such as authentication, sending messages, storing feedback, and managing notices and timetables. All the data is stored in a PostgreSQL database in an organized manner.

At the end, the project results in a complete and working system that not only makes communication easier but also improves efficiency of academic platforms.

Progress Overview

Most of the project work has been completed. Important features like login system, messaging, feedback, notice board, and timetable management are already working properly.

The frontend and backend are connected, and the system can handle basic user actions.

The project is going correctly, and our project is successfully working also it passed all the test cases, further the advanced features will be considered as our future work.

Codebase Information

i) Repository Structure:
Separate modules for:

- Frontend (React-based UI)
- Backend (Node.js + Express APIs)
- Database (PostgreSQL tables and queries)
- Authentication System (JWT login and role-based access)
- Messaging System
- Feedback System
- Notice Board
- Timetable Management
- Sentiment Analysis

Repository link: https://gitlab.com/dbms-v-t160/edubridge-web-dev-project.git

ii) Key Commits:

- Initial project setup and folder structure creation
- Database design and creation of tables
- Implementation of user login and authentication system
- Development of backend APIs for core features
- Creation of frontend pages using React
- Integration of frontend with backend using APIs
- Implementation of messaging, feedback, and notice features
- UI improvements and responsive design updates

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
API Endpoint Testing	Pass	APIs tested using Postman
Authentication Flow	Pass	Login and Logout working properly
Database Operation	Pass	Data stored and retrieved correctly
User Registration	Pass	Duplicate email check working
Messaging System	Pass	Messages sent and received correctly
Feedback Submission	Pass	Feedback saved properly
Faculty Search	Pass	Search gives correct results
File Upload	Pass	Files uploaded within size limits
Notice Board	Pass	Notices shown correctly
Security Testing	Pass	Data is secure and protected
Sentiment Analysis	Pass	Integration completed
Performance Testing	Pass	System works fast and smoothly

Deliverables Progress

- Login and authentication system → Completed
- Messaging system → Completed
- Feedback system → Completed
- Notice board → Completed
- Timetable management → Completed
- Frontend UI → Completed
- Sentiment analysis → Completed
- Testing → Completed